

Does Netflix Use Embeddings & Semantic Search?

A concise reference on how Netflix applies embedding models and vector similarity to power its world-class recommendation engine.

Q: Does Netflix actually use embeddings for recommendations?

Yes — embeddings are one of the core techniques behind Netflix's recommendation system. Both users and content are converted into dense numeric vectors (embeddings) and placed into a shared high-dimensional vector space. Recommendations then reduce to a fast nearest-neighbour search: find the content vectors closest to a given user's vector.

What Gets Embedded?

What	Signals used to build the embedding
Users	Watch history, ratings, re-watches, time-of-day, skip patterns, thumbs up/down, how long they browsed b
Movies & Shows	Genre, cast, director, plot synopsis, visual style, pacing, mood, and — crucially — how similar users inter

How the Recommendation Works

At inference time, the system takes a user's embedding vector and runs an **Approximate Nearest Neighbour (ANN) search** over millions of content vectors to find the closest matches. Cosine similarity is the typical distance metric — a small angle between two vectors means high semantic similarity.

- **Collaborative filtering** — Users with similar watch histories land near each other in the vector space. Their unwatched content surfaces as recommendations for you.
- **Content-based matching** — Two shows can be semantically similar even with no cast or genre overlap (e.g., "dark slow-burn thrillers") because their embeddings capture tone, pacing, and mood.
- **Semantic search** — When you type "feel-good 90s movie," Netflix embeds the query and matches it against content embeddings, not just keyword tags.
- **"Because you watched X" rows** — Take the embedding of X, find the K nearest content vectors, surface those titles.
- **Cross-language discovery** — A fan of Korean dramas gets recommended similar Japanese or Turkish shows because they occupy nearby regions in the shared vector

space, regardless of language.

Scale & Real-Time Performance

Exact cosine similarity over millions of vectors would be too slow. Netflix uses **Approximate Nearest Neighbour (ANN)** algorithms (e.g., libraries like **Faiss** from Meta) to retrieve the top-K closest vectors in milliseconds. The slight loss in exactness is acceptable — the recommendations are still highly relevant, and the system responds in real time as you browse.

Key Insight

Embeddings allow Netflix to bridge the gap between **what you watch** and **what you might enjoy** — even across genres, languages, and formats you've never tried before. A user who loves tense legal dramas might be recommended a thriller documentary purely because their vectors sit close together, with no explicit genre tag linking them.

Quick Reference Summary

Technique	What it does
User embeddings	Represent taste as a vector from watch/rating history
Content embeddings	Represent shows/movies by style, tone, genre, interaction data
Cosine similarity	Measures angle between vectors — small angle = similar taste
ANN search (e.g. Faiss)	Finds nearest content to a user vector in milliseconds
Semantic search	Matches typed query embeddings to content embeddings
Cross-lingual recs	Language-agnostic similarity via shared vector space

Source: Netflix Tech Blog (netflixtechblog.com) — publicly documented in research on their recommendation and personalisation systems.